

Yajat Mehta

+1 (623) 273 4531 | official.yajatmehta@gmail.com | [LinkedIn](#) | Tempe, AZ

EDUCATION

Arizona State University

B.S., Business Data Analytics | *B.S., Artificial Intelligence in Business* | *GPA: 3.56*

August 2022 – May 2026

Tempe, AZ

SKILLS

Analytics & BI: Tableau, Power BI, Advanced Excel, PivotTables, Dashboarding, Data Visualization, Data Mining, Data Governance

Languages & Query: SQL, Python, JavaScript, HTML, CSS

Data & ML: Pandas, NumPy, Scikit-learn, NLP, Word2Vec, Random Forest, Logistic Regression, Text Classification

Cloud & Big Data: Google Cloud Platform, Google Cloud Storage, PySpark, Spark ML

Systems & Modeling: Dimensional Modeling, Star Schema, Snowflake Schema, Data Warehousing, ERDs, Schema Design, SDLC

Tools & Databases: PostgreSQL, Supabase, Git, Docker, Postman, VS Code, Google Colab

Certifications: The Global Career Accelerator: Tableau, Advanced Excel; AWS: Machine Learning for Natural Language Processing; AWS Academy: Machine Learning Foundations

WORK EXPERIENCE

Desk Aide

August 2023 – February 2026

Arizona State University | *Documentation, Operations, Data Management*

Tempe, AZ

- Submitted and tracked maintenance requests, housing records, and administrative documentation while following **FERPA**-related confidentiality standards.
- Managed front-desk operations including service request intake, visitor logging, and resident inquiry resolution for a high-volume university housing environment.

PROJECTS

Medicaid Pharmaceutical Spending Analysis

May 2026

- Analyzed Medicaid drug reimbursement and 2024 population datasets to identify high-cost drugs, reimbursement pressure, and state-level spending variation across the U.S.
- Built **Tableau** dashboards with calculated per-capita reimbursement fields to compare drug costs, Medicaid vs. non-Medicaid spending, and geographic cost concentration.
- Determined that high-cost specialty drugs, rather than high-volume generics, drove the strongest reimbursement pressure — supporting targeted rebate negotiation and prior authorization strategies.
- Identified high-spending states and regional patterns to support state-level cost-control policy benchmarking.

Snowflake Schema Design for Ticket Sales Data Warehouse

December 2025

- Converted a transaction-grained star schema into a normalized **snowflake schema** by decomposing dimensional hierarchies while preserving fact-to-dimension relationships.
- Designed **SQL CREATE TABLE** statements with surrogate keys, primary keys, foreign keys, and hierarchical dimension relationships for a ticket sales analytical environment.
- Modeled a central fact table connected to date, attendee, event, venue, city, month, and year dimensions to enable structured multi-level analytical querying.

Sentiment Analysis Using GCP and PySpark

May 2025

- Built a scalable **PySpark** NLP pipeline on **Google Cloud Platform** to classify Amazon review sentiment across a large text dataset.
- Applied tokenization, stopword removal, **Word2Vec** vectorization, and **Random Forest** classification; evaluated results using F1 score with and without punctuation removal.
- Loaded source data from **Google Cloud Storage** and persisted the trained Spark ML model back to GCP for reuse — demonstrating end-to-end cloud data pipeline management.

Investigating Potential Lending Bias in Mortgage Applications — HMDA Data

April 2024

- Analyzed 12,550 HMDA loan applications to surface patterns of potential lending bias by race and ethnicity in mortgage approval rates using **Excel** and **PivotTables**.
- Conducted a statistical A/B test revealing a 22% higher approval rate for middle-class White applicants versus Black applicants, providing quantified evidence of potential systemic bias.
- Built Excel dashboards displaying approval/denial breakdowns (82.16% approved, 17.84% denied) segmented by demographic group for clear executive-level communication.

Data Center Location Selection Using Energy Surplus Data

April 2024

- Designed interactive **Tableau** dashboards analyzing energy surplus and renewable share across 11 U.S. regions to support data center site selection decisions.
- Built a dual-axis supply and demand chart identifying August as the peak energy surplus month for the Mid-Atlantic region and 3:00 PM as the top wind energy production hour in California.